

2025 UT Invitational

October 25, 2025

Water Quality B/C Exam



Parameters

- Each team may bring one 8.5" x 11" sheet of paper, and two class II calculators.
- Additional information here, such as lab / information specific to the event.

Directions

- You will have **50 minutes** to complete this exam - do not begin until instructed to do so!
- All answers must be indicated on your answer document. **DO NOT WRITE ON THIS EXAM!**
- Point values are indicated for all questions & tiebreakers are indicated in test booklets. The number of tiebreakers correctly answered indicates your tie order.
- The maximum number of obtainable points on this exam is 175.
- You may take apart the test to split the work, but make sure to staple it back together at the end of the event.
- Record your salinometer result on the last page. You're welcome to go to the salinometer at any time when it is not in use.

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(any questions after the conclusion of this event may be emailed to anebriitt@utexas.edu)

CLASS SET: DO NOT WRITE ON THE EXAM

1. (1 point) Freshwater makes up what percentage of total global water?
 - a) 0.3%
 - b) 1.0%
 - c) 2.5%
 - d) 10%

2. (1 point) Which of these are used as indicator organisms in the bioassessment of streams?
 - a) Phytoplankton
 - b) Zooplankton
 - c) Periphyton
 - d) Neuston

3. (1 point) Which of the following correctly describes the dominant functional feeding group for headwaters, mid reaches and large rivers, according to the RCC?
 - a) Shredders, collectors, predators
 - b) Shredders, scrapers, collectors
 - c) Collectors, scrapers, predators
 - d) Scrapers, predators, collectors

4. (1 point) A shaded, narrow headwater reach has coarse leaf packs, an abundance of stonefly shredders, and few periphyton mats. The dominant energy source is:
 - a) Sunlight to periphyton
 - b) FPOM from upstream
 - c) CPOM from riparian inputs
 - d) Runoff

5. (1 point) The number of species in a sample collected from a stream is referred to as
 - a) Richness
 - b) Density
 - c) Diversity
 - d) Abundance
 - e) None of the above

6. (1 point) In a newly formed forest gap, a dense cohort of maple saplings is established. As the canopy closes, taller stems overshadow neighboring saplings, which show reduced growth and begin to thin out. This interaction is best described as:
 - a) Interspecific competition
 - b) Intraspecific competition

- c) Competitive exclusion
 - d) Resource partitioning
7. (1 point) The predicted P/R(photosynthesis/respiration) ratio in shaded headwaters is:
- a) >1
 - b) <1
 - c) ≈ 1
 - d) Not predictable
8. (1 point) Which setting is most likely to support an extensive hyporheic zone with high insect diversity?
- a) Straight, bedrock-confined channel
 - b) Meandering channel with fine sand bed
 - c) Meandering channel with gravel/cobble bed
 - d) Large, canyon-bound river
9. (1 point) Which pairing is the most accurate given usual stream habitats?
- a) Many aquatic insects $\rightarrow$ riffles
 - b) Salmonid spawning $\rightarrow$ pools
 - c) Fish resting from current $\rightarrow$ runs
 - d) Low dissolved oxygen $\rightarrow$ riffles
10. (1 point) Along a meander, cobble on the inner bank are clean and building upward; the outer bank shows fresh silt lenses and toppled saplings. What describes what's happening on each bank?
- a) Erosion inner, deposition outer
 - b) Deposition inner, erosion outer
 - c) Equal erosion both sides
 - d) Debris-flow emplacement inner, aeolian deposition outer
 - e) None of the above
11. (1 point) In lakes, most settling detritus is decomposed primarily in the:
- a) Epilimnion
 - b) Limnetic zone
 - c) Littoral zone
 - d) Profundal zone
12. (1 point) Which trophic state typically has cold, clear water and high hypolimnetic oxygen
- a) Oligotrophic

- b) Mesotrophic
- c) Eutrophic
- d) Hypereutrophic

13. (1 point each) Match the metric to the most likely trophic state:

a) Oligotrophic b) Mesotrophic c) Eutrophic

- i. Chlorophyll-a ~ 30 µg/L ___
- ii. Secchi depth ~ 10 m ___
- iii. Total phosphorus ~ 20µg/L ___

14. (1 point) Lake Okeechobee is known to be an eutrophic lake that often can appear murky in certain areas. Which of the following is the most direct cause?

- a) Phytoplankton populations
- b) Wind/wave resuspension of bottom sediments
- c) Pollen from vegetation surrounding shoreline
- d) Human activity

15. (1 point) Which of these is NOT a non-point source of pollution?

- a) Landfills and spills
- b) Agricultural runoff
- c) Septic systems
- d) Municipal wastewater treatment plants

16. (1 point) The U.S. Geological Survey divides bodies of water based on the amount of dissolved solids. Which of these would classify it as a body of freshwater?

- a) 900 ppm
- b) 2500 ppm
- c) 3200 ppm
- d) 10000 ppm

17. (1 point) High total coliform counts in drinking water most directly indicate increased risk of which health outcome?

- a) Diarrhea
- b) Cholera
- c) Hepatitis A
- d) Giardiasis

18. (1 point) If flow velocity doubles, competence (maximum grain size movable) changes by approximately what factor?
- a) 2x
 - b) 4x
 - c) 8x
 - d) 64x
19. (1 point) Youthful Stream Valleys are NOT associated with which of the following features?
- a) V-shaped valley
 - b) Ungraded
 - c) Flood plain
 - d) High sediment load
20. (1 point) Which statement best describes how dissolved oxygen and water temperature typically differ in small streams draining forested vs. urban catchments?
- a) DO higher in forested and temperature lower in forested
 - b) DO higher in urban and temperature lower in urban
 - c) Both DO and temp higher in forested
 - d) Both DO and temp higher in urban
21. (1 point) Which of these is a nonnative species in Texas?
- a) Common carp
 - b) Bluegill sunfish
 - c) Largemouth bass
 - d) Guadalupe bass
22. (1 point) Which of these is NOT a type of erosion in streams?
- a) Hydraulic action
 - b) Abrasion
 - c) Chemical solution
 - d) Wind erosion
23. (1 point) In a channel, flow velocity primarily depends on which factors (select all that apply): (no partial credit, all or nothing)
- a) Roughness
 - b) Turbidity
 - c) Depth of channel
 - d) Slope

24. (1 point) In an ideally graded river, why can average velocity stay similar or even slightly increase downstream even though slope decreases?

- a) Channel straightening downstream speeds the flow
- b) Larger discharge and hydraulic radius reduce frictional losses
- c) Grain size always gets finer downstream
- d) Coriolis forces accelerates lowland rivers

25. (1 point) Continuous sensors show strong diel swings under steady flow. This pattern most strongly indicates:

- a) Diurnal variation in wastewater discharge volume
- b) Barometric pressure cycles affecting probe readings
- c) High primary productivity
- d) Temperature fluctuations

26. (1 point) What two biodiversity factors does Simpson's index take into account?

- a) Biomass and primary productivity
- b) Habitat heterogeneity and area
- c) Population size and growth rate
- d) Species richness and species evenness

27. (1 point) Which of the following rocks would likely make a good aquifer?

- a) Sandstone
- b) Granite
- c) Gneiss
- d) Conglomerate

28. (1 point) Vinegar has a pH of 3 while milk has a pH of 6. Which of the following is true?

- a) Vinegar has 3 times more H^+ than milk
- b) Milk has 1,000 times more H^+ than vinegar
- c) Vinegar has 1,000 times more H^+ than milk
- d) Vinegar has 30 times more H^+ than milk
- e) Vinegar has 100 times more H^+ than milk

(1 point each) *Refer to Image 1 for questions #29-#31*

29. What is the stream order at point A?

30. What is the stream order at point B?

31. What is the stream order at point C?

Two small stream sites were sampled in late spring. The following was logged:

Metric	Site A	Site B
Total taxa richness	42	28
EPT richness	18	5
%EPT	46%	14%
Tolerant-taxa richness	4	9

32. (1 point) Which is best supported by the data?

- a) Site A is more impaired -> high EPT indicates organic enrichment
- b) Site B is likely more impaired -> low EPT and increased tolerant taxa suggest poor water quality
- c) Sites are equivalent -> EPT richness depends only on total richness
- d) Site B has better habitat complexity as the EPT is lower

(1 point each) For questions #33 - #39, answer either with true or false for the following statements.

33. Phosphorus is the most limiting nutrient on land, while nitrogen is the most limiting nutrient in freshwater.

34. Sample A has 10 species and sample B has 20 species. All species in both samples have 20 individuals (i.e. both samples have the same evenness). Sample A has a higher Shannon-Wiener Diversity than sample B.

35. Leaves of aquatic macrophytes decompose faster than do the leaves of terrestrial plants.

36. The total number of benthic species you would find in a sample of 1 m² of the stream bed is likely to be the same as the number of species you would find in a sample of 10 m² randomly collected individuals.

37. An increase in the amount of discharge to surface water produces a lower water table.

38. Under sunlight, high-discharge, fast-moving streams warm less from the sun than small, slow streams.

39. As turbidity increases, Secchi depth generally increases.

40. TIEBREAKER #1 (5 points) You titrate two 10.00 mL aliquots of fixed stream water using a standard solution of 0.00100 M $\text{Na}_2\text{S}_2\text{O}_3$. Aliquot A (first titration) requires 6.40 mL sodium thiosulfate. Aliquot B (second titration) requires 2.00 mL. Calculate the dissolved oxygen concentrations in mg/L. (Hint: this question is related to the Winkler method)

41. (2 points) Based on the previous question, what percent dissolved oxygen was consumed? Round to one decimal place.

42. (5 points) You're working on a field project at UT collecting samples near the stadium at Waller Creek for a BOD_5 test. Back in the lab, you notice that there aren't enough microorganisms, so you decide to run a seeded BOD_5 test. You prepare a 300 mL BOD_5 bottle with 20.0 mL of creek sample and fill the rest with seeded dilution water. The dilution water used for sample bottles is prepared at 4.0 mL seed per liter. On Day 0, the diluted sample's dissolved oxygen is 8.9 mg/L. After 5 days of incubation, the dissolved oxygen is 2.6 mg/L. To correct for the seed's oxygen uptake, you also run a seed control using dilution water seeded at 10.0 mL seed per liter. Its dissolved oxygen readings are $B_0 = 8.7 \text{ mg/L}$ and $B_5 = 7.9 \text{ mg/L}$. Calculate BOD_5 in mg/L for the creek sample.

43. (2 points) Interpret this result. What does this indicate about Waller Creek and the outcome it will likely have on aquatic life?

44. (1 point) A standard BOD_5 test is incubated at what temperature?

For questions #45-49 refer to the following scenario:

A spring-fed Texas Hill Country headwater supports $N_0 = 12,000$ trout. Under a co-management agreement, the indigenous nation applies TEK (traditional ecological knowledge) to set a 6-week no-net window during spawning season in order to reduce adult and by-catch mortality.

Year 1 observations shows the following:

Births: 3,000 *Immigration:* 450

Deaths: 2,100 *Emigration:* 725

Of the deaths, 30% occur during the 6-week window.

45. (2 points) Find the crude birth rate (CBR) and crude death rate (CDR) per 1,000 fish for year 1.

46. (2 points) Find the net population change and the percent growth rate for Year 1.

47. (1 point) Estimate the doubling time of the population. Round to the nearest tenth (Hint: use the rule of 70).

48. (4 points) A TEK-informed seasonal closure reduces mortality during spawning season by 20%. Calculate the (i) deaths, (ii) net change, (iii) growth rate and (iv) doubling time for Year 1 with this closure.

i.

ii.

iii.

iv.

49. (1 point) If the TEK-adjusted growth rate from the previous question continues for 4 years (compounded annually), what is the expected population at the start of Year 5?

For questions #50-51, refer to the following scenario:

A researcher visits an area of interest and wants to estimate the total population of a type of fish.

They use traps to capture 190 fish. Each of these fish are marked with a numbered tag and released

back into the environment. On the second day, the researcher returns to the area and captures 250 fish where 60 are marked.

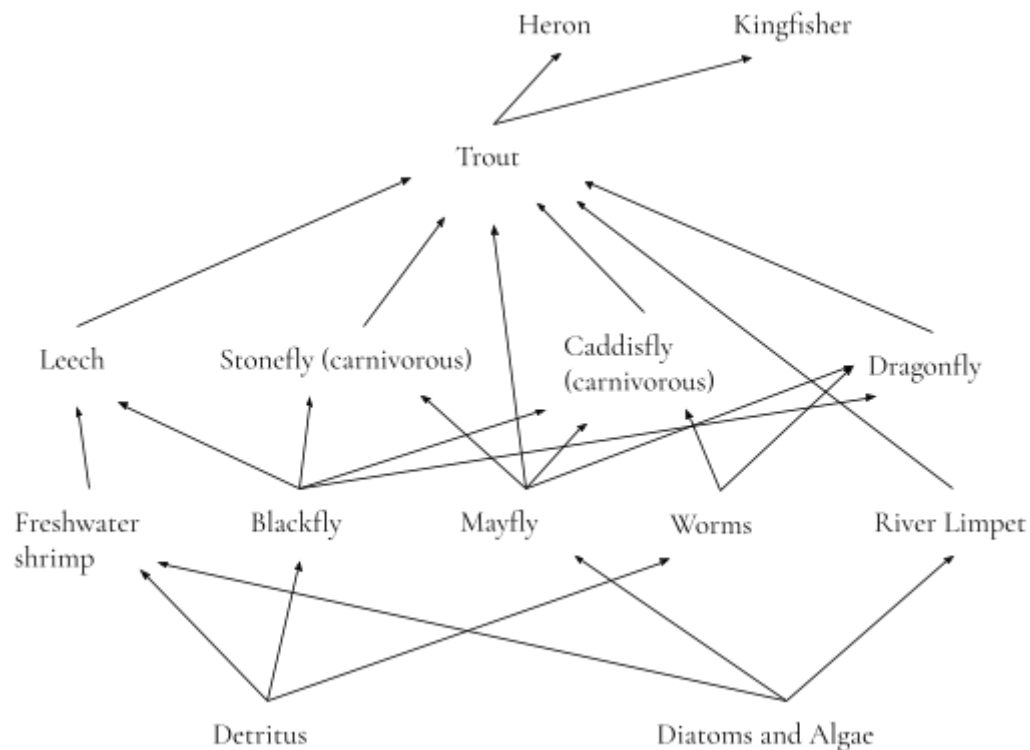
50. (2 points) Find the estimation for the total population to the nearest fish.

51. (1 point) Using your population estimate, what is the capture probability on the second day for a randomly chosen fish in the pond?

52. (1 point) Refer to Image 2, at 5 mm and 0.70 m/s, the Hjulstrom curve predicts:

- a) Deposition
- b) Transport
- c) Erosion

53. (2 points) Refer to Image 2, the shear stress method gave $\tau = 6.4$ Pa and $\tau_c = 4.25$. Does the Hjulstrom curve prediction at 5mm, 0.70m/s agree or disagree? Explain your reasoning.



Refer to the diagram above for questions #54-57

54. (1 point) Which removal is most likely to trigger a top-down trophic cascade on grazers of algae?

- a) Remove river limpets
- b) Remove mayflies
- c) Remove trout
- d) Remove diatoms & algae

55. (1 point) Which organism is most likely to experience the highest biomagnified concentration of a persistent pollutant if introduced at very low levels into producers?

- a) Blackfly
- b) Dragonfly
- c) Trout
- d) Heron

56. (1 point) In the following food chain, what order consumer is the trout?

Diatoms & Algae → Mayfly → Trout → Heron

- a) First
- b) Second
- c) Third
- d) Fourth
- e) None of the above

57. (1 point) According to the 10% rule, what is the maximum energy available to the trout population if primary producers capture 50,000 kJ d⁻¹?

- a) 50 kJ d⁻¹
- b) 500 kJ d⁻¹
- c) 5,000 kJ d⁻¹
- d) 500,000 kJ d⁻¹

58. (1 point) Refer to Image 3, what type of stream is depicted in the image?

(1 point each) *For questions #59 to #62, circle the correct word that best completes the statement about the type of stream you identified in question #.*

59. The bars in these types of streams are often (stable/unstable).

60. The channels have a (high/low) width-to-depth ratio.
61. This occurs when sediment load (exceeds/is less than) stream capacity.
62. The sediment is typically (well-sorted/poorly sorted).
63. (1 point) Deepest part of a river channel is called a:
64. (1 point) Refer to Image 4, what major US river is shown here flowing eastward from Texas to Louisiana? (Hint: the rivalry between UT and OU references this river)

You want to calculate the flow of a section of this river and measure the width of it to be 28.7 m and depth as 13.5 m. You then drop a rubber duck into the river and measure its time over a distance of 5.0 meters. Because the velocity of the water varies across the width of the stream, you repeat this process multiple times and create the following data table:

Trial #	Time(s)
1	4.1
2	3.8
3	4.5
4	3.9
5	4.7

65. (1 point) Calculate the average velocity to one decimal point.
66. (2 points) Calculate the discharge of the section of the river to one decimal point.
67. (6 points) Lake Minnetonka is a dimictic lake and thus has a seasonal cycle of thermal stratification. Plot how temperature changes with depth for each season and write in the thermal layers (i.e. epilimnion, metalimnion, hypolimnion or mixed/no layers when appropriate)

68. (2 points) Rivers and Streams have the natural ability to recover from the introduction of oxygen-demanding substances through physical, biological, and chemical processes. What would happen if a normally well-balanced stream suddenly received thousands of gallons of raw sewage?

69. (2 points) Weather conditions such as heat and sunlight can increase the temperature of water.

70. (1 point) Based on this, what do you expect to happen to fish populations during the Texas summer months?

71. (2 points) You're working at an Austin water-quality facility and observe hyper-excitable fish that exhibit rapid, erratic movements followed by death. A handful of them even attempt to swim onto shore. What are possible cause(s) of this behavior?

72. (2 points) A soil sample collected from a local stream was found to have high sand content. Is the water found in this stream likely to be more basic or acidic? Why?

73. (2 points) Lakes and other bodies of water are able to maintain a neutral pH through their surrounding soil. Write out the two balanced net-ionic chemical equations for how lakes underlain by limestone create a buffer for acidic rainfall.

1.

2.

74. (1 point) Which statement about the manometric BOD method is NOT true?

- a) It provides a continuous display of BOD
- b) It can give a direct BOD reading without titration
- c) It uses pressure change to infer oxygen uptake
- d) It requires dilution water and seeding for every sample

75. (1 point) Which treatment is most appropriate for removing nitrate from a small water system?
- a) Sand filtration
 - b) UV disinfection
 - c) Ion exchange or reverse osmosis
 - d) Aeration only
76. (1 point) What is the cutoff for TDS in drinking water in the US?
- a) 50 mg/L
 - b) 250 mg/L
 - c) 400 mg/L
 - d) 500 mg/L
77. (1 point) What does TSS stand for?
78. (1 point) Which U.S. law establishes federal wastewater (effluent) standards?
79. (1 point) Another name for Forchhammer's principle is:
80. (1 point) Which agency took over the primary enforcement authority of the Safe Drinking Water Act in Texas in 1978?
- a) Texas Water Development Board
 - b) Texas Commission on Environmental Quality
 - c) Texas Department of Health
 - d) Environmental Protection Agency
81. (1 point) While working on your field project, you need to sample slow backwaters and vegetated pools without wading too deep. Which of the following is the best choice for this?
- a) Dip net
 - b) Kick net
 - c) D-net
 - d) Surber sampler
82. (1 point) Regardless of gear, which mesh size should you use to meet the US EPA benthic macroinvertebrates standards?
- a) 2.0 mm
 - b) 1.0 mm
 - c) 0.6 mm

d) 0.2 mm

83. (1 point) After coagulation/flocculation, the next process is typically:

- a) Aeration
- b) Sedimentation
- c) Filtration through sand
- d) Sludge digestion

84. (1 point) What is the primary use of Sodium Hypochlorite in wastewater treatment processes?

- a) pH adjustment
- b) Nutrient removal
- c) Sludge thickening
- d) Disinfection of wastewater

85 (1 point) Which phase of municipal wastewater treatment mainly focuses on the removal of suspended solids and organic matter through sedimentation?

- a) Preliminary Treatment
- b) Primary Treatment
- c) Secondary Treatment
- d) Tertiary Treatment

86. (5 points) TIEBREAKER #2 On the graph below, draw the daily cycle for DO, CO₂, and pH on the axes provided (please label or use some indication on which line is which, otherwise we won't be able to grade it properly :!)

+1.5 for correct DO

+1.5 for correct CO₂

+2 for correct pH

Part II: Identification

(1 point each) Refer to Image 5 for questions #87- #93

87. What species is depicted here?

88. This organism produces both male and female gametes. What is the term for this?

89. Being acoelomates and having no specialized organs, how do they get oxygen?

90. What type of symmetry does this organism have?
91. How many germ layers do they have?
92. What is this organism's pollution tolerance level?
93. Because of this, what water quality does its presence often indicate?

(1 point each) *Refer to Image 6 for questions #94-#97*

94. This is the larvae for what organism?
95. What are the larvae commonly known as?
96. Why do people often seek out control measures even though they are considered good water quality indicators?
97. What type of metamorphosis does this organism have?

(1 point each) *Refer to Image 7 for questions #98 - #101*

98. What species is depicted in Image 7?
99. This organism's rapid circling behavior mainly helps them:
- a) Thermoregulate
 - b) Detect vibrations
 - c) Court mates
 - d) Avoid oxygen-rich areas
100. Defensive chemicals are released from:
- a) Antennae
 - b) Maxillary glands
 - c) Pygidial glands
 - d) Coxal glands

101. Eggs are attached to:

(1 point each) *Refer to Image 8 for questions #102 - #107*

102. What is the common name of this organism?

103. Where was this organism first introduced in the US?

104. This organism reproduces via _____

105. This organism attaches to hard surfaces through _____

106. True or false: This species generally decrease water clarity by stirring up sediments

107. This organism increases or decreases water clarity?

(1 point each) *Refer to Image 9 for questions #108-112*

108. What is the common name of this organism?

109. What type of metamorph is this organism?

110. This organism is often confused with what other freshwater macro invertebrate in the WQ rules?

111. What distinctive shape does this organism form when mating?

112. A key distinction between the two is through its wing position. At rest, what does this look like for each one?

(1 point each) *Refer to Image 10 for questions #113-117*

113. What is the common name of this organism?

114. What is a key characteristic to this species' morphology?

115. How was this species introduced to the U.S.?
116. How do males and females differ in diet?
117. This organism is what apart which functional feeding group?
- a) Collector-filterers/gatherers
 - b) Scrapers
 - c) Engulfers
 - d) Piercers

(1 point each) *Refer to Image 11 for questions #118-122*

118. What is the common name of this organism?
119. What is unique about its coloration compared to most insects?
120. Air is stored under the _____ for respiration
121. Which legs are enlarged into oar-like paddles for swimming?
122. The diet of this organism consist of (only have to mention one for credit):

(1 point each) *Refer to Image 12 for questions #123-127*

123. What is the common name of this organism?
124. Some larvae species of this organism appear red. What gives them that color?
125. What advantage does this give them?
126. The adult approximate adult body length is:
127. What notable adult behavior do these insects exhibit near shorelines and lights?

(1 point each) *Refer to Image 13 for questions #128-132*

128. This is the larvae for what organism?

129. What water quality does its presence often indicate?

130. What is the feeding behavior of this organism as an adult?

131. What bioassessment index are they apart of?

132. What other two organisms are apart of this index?

TIEBREAKER #3

133. (10 points) Salinometer Result: ___